

A Packing of 7 Four-Spaces in $\mathbb{R}^{13}$

Limin Ge*

October 9, 2026

Abstract

We give 7 4-dimensional subspaces of $\mathbb{R}^{13}$ whose smallest squared chordal distance is at least 3.230536. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 3.164606, obtained by deleting 1 of the 8 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 7 subspaces of dimension 4 in $\mathbb{R}^{13}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 3.230536 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 13$, $n = 4$. The cell $N = 7$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 3.164606, obtained by deleting 1 of the 8 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(13 - 4)/13 \cdot 7/6 = 3.230769$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = Y G^{-1} Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n4N7.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
323053609839238926561007369608448533212574715874650179943733174261541000
069547042728907043866649322696194113069271438979706471736510721620601646
5771709869530930713696465590903468058753163835199
```

denominator:

```
100000000000044830294694032275104071878498968183775351025635314361140033
393766188052347495466096032832633720162188693968472201353081223093706393
9457043760389704603355279207589776330525025387904
```

B Coordinates

7 blocks of 13 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m13n4N7.txt`.

```
300718984965 259976131252 489662981899 221087530107-352085348624 477396539140 311476041222 179705832594194063853974 -327436085998 17078647832 -578296614175
-425283096101 -71976754916 382889777168 -335044225921905240527 398284594526 -207396885221 -2069529707345899952566 387108826345 -185357369917 -83046800882
-67056567013 -24783531503 421130084701 -70718171161-37552366910 461386463362 25832075363 243494836314 155130825399 122226179863 -628635318773 49566487855
124556440331 156564141075 79634063748 352912188068 -27399194681 44345371609 17051330919 -598293379873 -288147294983 169599599260 -165777269901 -183297742568
36593682571 -243173695207 -253989063548 286645108574177469033347 -197676238085 -165878652348 -3573540855816273985 -164431064415 113564925983 -166458141007
246113160446 8659649642 47046161550 -101160135603 -47462781405 321656945783 -542147164471 -273010784993228896027660 42780828530 13836911924 403957402141
-438797958420 391240925010 -52019772541 -3124383924735620610891 -15059549870 67612762342 -26937840419452435592424 -391194008623 -13214967000 283765300197
-231518106396 30608481986 7614596254 293512042108 -437147969789 -9714873428 -374103462274 73384567773-230200501248 -465569300201 67098882804 -238089278252
-39873949757 -417809027890 -184624492052 -702858442735182494800 -57588969690 -213122453131 53705315616559695649061 211955193238 298976545071 -52042350869
-536714247765 -196389055475 291941320091 508493852733011172446 424344891602 -146307641423 392423887736-315412269460 -84992265301 -307261270377 166392125126
191461768516 -408535589084 332113080149 -386026513056180477416 -318508165164 -25047272332 2179600463527150419929 -443259152114 -257688572831 213047579653
-102566883652 497185551872 -205036628573 -1221535405300708318903 334848542241 -548844764835 -2235247654535789224490 -144318743392 72594168862 215698383831
252599946448 248319517249 297977467796 75549221669 42848547127 210336372830 337347535533 311245524109 76565989895 184714869555 -602051693449 -376147570278
189540401858 -124601682371 64529451113 -4749061932509495226755 43113577472 -166047376968 -43738105327313698883483 -237028932849 5562300301 195390387656
438452704003 243721875884 33004746229 367157042960 -217537944373 98454842400 609659677005 -2158501212076705915037 -460077826223 -24426691291 -533388840431
68878652044 -399017358042 721199881294 -1524265946400552805816 55134620057 -91870593909 -118535842348-295943077691 -33933863201 579818012913 -227650893301
-108014036535 2732997419 61237704920 655160726733 -321270252269 -119878147004 2420840369 -31147317924717707611301 433024540497 533360942932 -87469502931
-512690769798 -239400062196 -106941519328 336187990732338686322 116274579099 -2012881001 -81677228694 -476973996207 1864488311 -275836509332 -321954993682
227262824362 47498161936 -15874480559 248998200040 -121700746509 -97570273650 -293303372684 115262075927585015721 -482120275660 -104067710434 257006550277
63570939888 217139281865 210964686614 393115707176 -233182572015 -640922377676 -146360799912 165017123237400503888 -169660937580 192856492894 200549276360
227558149388 440684660297 115368161137 -10582571429390133434204 309868638339 -48543475760 -84838926045878206475616 -39863906871 230330774839 -77197277908
540174903984 -286016095950 -425819402262 554916138465080991198 517426637906 297088249755 -186222819927-333958205478 172307769475 55572909995 -17251904182
-62497161098 -111023180185 -252599016277 2856728520767762612905 42956459582 450447905098 225167344338 -230152274095 16240202669 472284796754 48692211082
-291791337687 550999009442 -115987777889 -7162339717353100888 -41116509156 37188421641 -48633939374 314105140107 -205051961996 -36495546198 -564946200297
-9565823126 231959226958 179837864886 42950319448 140898925504 135162988076 119466810039 21671338931344055258338 175173878272 -340551640090 -334880282043
-64572883900 -115734605445 -332324635648 -32372550309636688235 6314763755 207432038025 -558264118346 349115168049 111169196126 -110505794304 274732200719
134489569129 345880134446 -237189083732 27721601635529314337661 475726021293 -30507224980 -5605897195640136010930 80233640770 3895740938 293631625142
-259885927077 -307553371980 -201228820505 5942026634222689116682 74535499163 -271120666348 220214210416494834972160 215214946093 -390168992549 425029550852
-413262388214 68949560947 256904689868 -49553112345717173341448 -42111497809 -309091131225 -23714411494921396687034 -617958687409 174519831197 11715997964
281314974226 -90128488415 156813555848 -4189731759333839575065 449207008936 -55638363714 -255070504672
-153718902806 169287803880 453546901389 -1788451989067786254851 -54653124939 -217371683131 -192135803878
```